

ASSIGNMENT – 1

Introduction to Sensors and Actuators (UEC 3942) – Question 16

AIM

To design a structural health monitoring (SHM) system for a bridge using resistive strain gauges that continuously measures stress and detects structural deformation, including sensor placement, signal conditioning, data processing, and damage-detection logic.

PROBLEM STATEMENT

Bridges deteriorate silently: corrosion, fatigue cracking, cable loosening, and overload events reduce stiffness long before any visible damage. A monitoring system must measure the live strain at the bridge's critical sections around the clock, separate real load effects from temperature effects, and raise an inspection alert when the structure's behaviour drifts from its healthy baseline.

SYSTEM OVERVIEW

Each measurement node combines strain gauges in a temperature-compensated Wheatstone bridge, an instrumentation amplifier and 24-bit ADC, and a microcontroller that converts readings to stress, runs FFT analysis on the dynamic component, and checks alarm limits. Nodes report over a Zigbee mesh to a gateway with cellular backhaul.

Fig 1. Bridge SHM - complete measurement chain

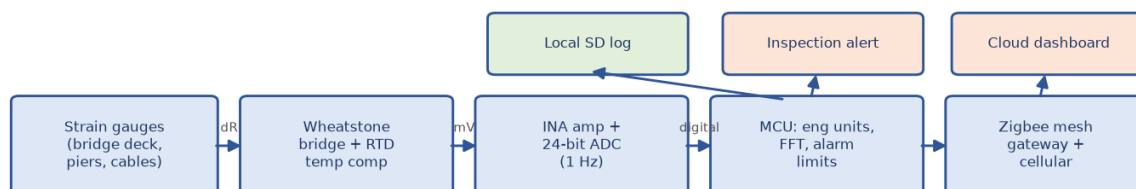
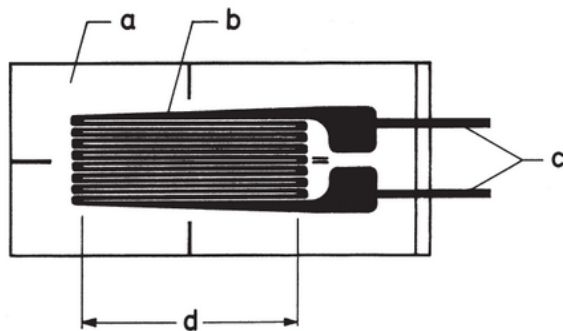


Fig 1. Complete measurement chain of the bridge SHM node

SENSING PRINCIPLE — RESISTIVE STRAIN GAUGE

A metallic foil strain gauge changes resistance when deformed: $R = R_0(1 + GF \cdot \epsilon)$, with a gauge factor $GF \approx 2$ for constantan foil. Steel structures work up to $\sim 2000 \mu\epsilon$, so the resistance change is tiny (350 gauge $\sim 0.7\%$ at full scale). Four gauges wired in a Wheatstone bridge (two in tension, two in compression) multiply sensitivity by four and cancel axial/bending-unwanted components; the bridge output is only a few millivolts, which sets the amplifier requirement.

Paul Eisler in 1952, imitating printed circuit techniques, made a strain gauge, with the geometric shape currently used to make metal foil resistors.



- a carrier material
- b measuring grid
- c connections
- d effective grid length

Foil strain gauge – the sensing element bonded to the structure

- Gauge: constantan foil, 350 (CEA-series), with self-temperature-compensation matched to the steel's expansion coefficient.
- Bonding: surface abraded, cleaned, bonded with cyanoacrylate/epoxy, coated with waterproofing + mechanical guard – installation quality decides everything.
- Bridge: half-bridge (bending) or full-bridge configuration; shunt calibration resistor verifies the gain periodically.
- Why strain gauges over other sensors: cheap per point, proven for decades on steel and concrete, direct stress measurement, DC-coupled so both static (dead load) and dynamic (traffic) strain are captured.

SENSOR PLACEMENT ON THE BRIDGE

Gauges go where the stress analysis says the extremes live: midspan bottom fibres (maximum positive bending/tension), top fibres over the piers (maximum negative bending/compression), and on main cables or truss members in cable/steel bridges. A co-located RTD measures temperature so thermal expansion (= LT) can be subtracted analytically from the strain reading.

Fig 2. Strain gauge placement on the bridge (tension / compression zones)

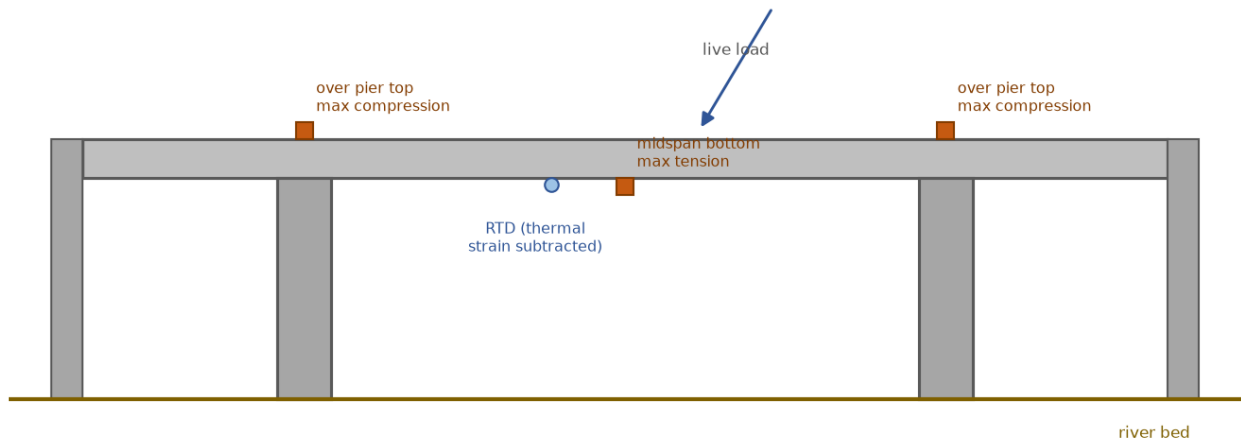


Fig 2. Strain gauge placement: midspan tension, pier-top compression, co-located RTD

SIGNAL CONDITIONING AND ACQUISITION

- Bridge excitation: 5 V DC, low-drift reference; sense wires (6-wire Kelvin connection) cancel cable-resistance errors over long runs to the junction box.
- Amplification: INA125/INA128 instrumentation amplifier, gain 100–500, raises the millivolt bridge output to the ADC range; RC anti-alias filter before sampling.
- Conversion: 24-bit sigma-delta ADC (ADS1256-class) at 1 Hz for quasi-static strain, with a faster burst window (50–100 Hz) for dynamic/FFT analysis.
- Cabling: shielded twisted pair to a weatherproof junction box; surge protection for outdoor runs.
- Power: solar panel + LiFePO4 battery per node; duty-cycled acquisition gives year-round unattended operation.

PROCESSING AND DAMAGE DETECTION

The microcontroller applies three complementary checks to the strain history: static overload, abnormal strain rate, and modal-frequency drift.

Fig 3. Natural-frequency drift from FFT of dynamic strain = stiffness-loss indicator

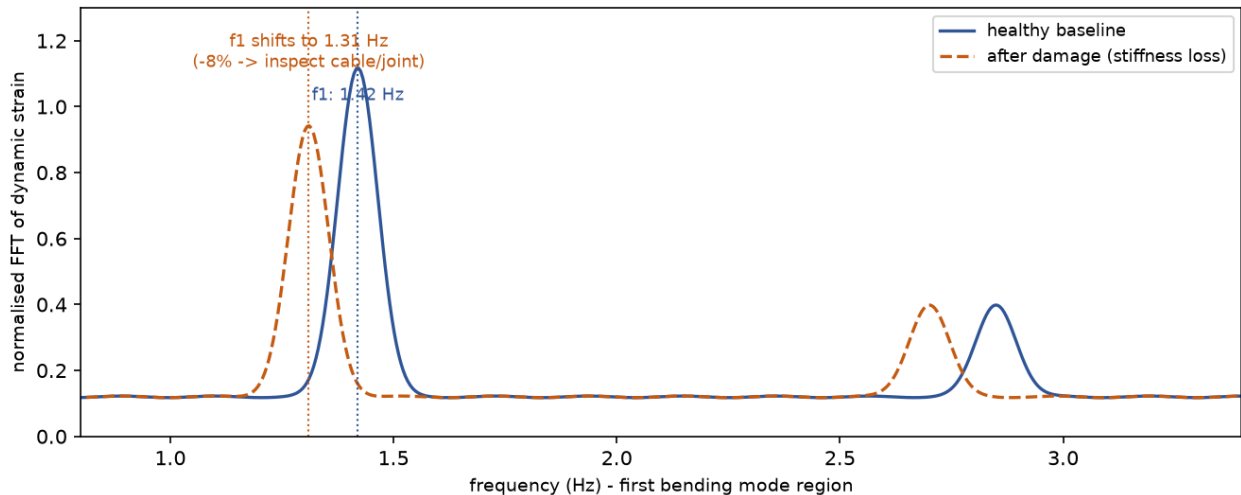


Fig 3. FFT of dynamic strain: natural-frequency shift reveals stiffness loss

- Static check: convert strain to stress ($\sigma = E \epsilon$, $E \approx 200$ GPa for steel); inspect-alert if quasi-static strain exceeds 2000μ (0.2%).
- Rate check: a sudden strain-rate spike indicates an impact or overload event — logged with timestamp for correlation with permits/accidents.
- Modal check (the clever one): FFT of the dynamic strain under normal traffic extracts the natural frequencies. Stiffness loss from cracking or cable loosening shifts natural frequencies down — an 8% drift in the first bending mode (e.g. 1.42 Hz $\rightarrow$ 1.31 Hz) triggers an inspection, long before visible damage.
- Baseline statistics: healthy-frequency fingerprints stored per season (temperature also shifts frequency slightly), compared with control charts so only statistically significant changes alert.
- Data pipeline: edge MCU computes features and alarms locally; raw/featured data flows over 802.15.4 Zigbee mesh to a gateway with cellular backhaul; local SD card buffers during network outages.

COMPONENT SELECTION

Block	Component (example)	Key specification	Why chosen
Strain sensor	CEA-06-250UW-350 foil gauge	350 , GF $\approx$ 2.05, STC matched	Proven, cheap, direct stress measurement
Bridge network	4-gauge full/half Wheatstone	6-wire Kelvin, shunt cal	4 $\times$ sensitivity, temperature and bending compensation
Temperature ref	PT100 RTD co-located	± 0.1 °C	Subtract thermal strain analytically
Amplifier	INA125 instrumentation amp	Gain 100–500, low drift	Resolves μ V-level bridge outputs

Block	Component (example)	Key specification	Why chosen
ADC	ADS1256 24-bit sigma-delta	2000 S/s max, 23-bit noise-free	Resolves microstrain over years of drift
Controller	STM32L0 low-power MCU	FFT in burst mode, <10 μ A sleep	Edge features + alarms, solar-powered
Telemetry	802.15.4 Zigbee mesh + cellular GW	Mesh self-healing across span	Covers the whole bridge without cabling
Power	10 W solar + LiFePO4	5-day autonomy	Unattended year-round operation

PERFORMANCE AND PRACTICAL CONSIDERATIONS

- Resolution: 24-bit chain resolves $< 1 \mu$ — enough to see daily thermal cycles and individual vehicle passages.
- Alarm policy: strain $> 2000 \mu$, strain-rate spike, or statistically significant modal-frequency drift inspection alert with node ID and data snapshot.
- Environmental: gauges waterproofed and guarded; junction boxes IP67; lightning/surge protection on all outdoor cabling.
- Installation practice: gauge installation is the weakest link — surface prep, alignment, and cure time are quality-controlled; spare channels installed for redundancy.
- Scalability: same node design scales from a single span to hundreds of channels; the mesh adds nodes without new cabling.

LEARNING OUTCOMES

- Understood the strain-gauge principle ($R = R_0(1 + GF \cdot \epsilon)$) and Wheatstone bridge configurations for sensitivity and compensation.
- Learned where gauges belong on a bridge and why temperature compensation with a co-located RTD is essential.
- Designed the conditioning chain: Kelvin excitation, instrumentation amplifier, 24-bit sigma-delta ADC.
- Applied FFT-based natural-frequency monitoring as a stiffness-loss/damage indicator with statistical baselines.

REFERENCES

- D. Patranabis, Sensors and Transducers, PHI Learning.
- J. Fraden, Handbook of Modern Sensors, Springer — strain gauge chapter.
- J. M. Ko and Y. Q. Ni, 'Technology developments in structural health monitoring of large-scale bridges', Engineering Structures, 2005.
- UEC 3942 Introduction to Sensors and Actuators, SSN College of Engineering, Regulations 2024.

